Wire materials and processing

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Lecture Production of electric motors and machines

The lecture unit "Wire materials and their processing" provides an overview of the main conductor materials, as well as their manufacture and processing.

Conductive windings are one of the main components of electrical devices, and have a significant influence on machine performance, efficiency, and machine size.

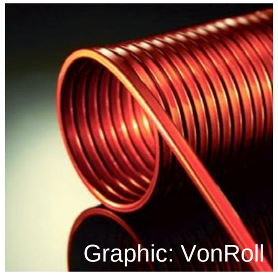
After attending the lecture unit "Wire materials and their processing", students shall:

- **be able to differentiate between the most important conductor materials and geometries,**
- **understand the relevance of the fill factor of a winding,**
- **be able to reproduce the process chain for manufacturing electrical conductors,**
- **understand the refinement processes of the conductor materials and**
- **be able to explain the relevant standards and examination modalities for the qualification of conductors.**


Graphic: Essex


Graphic: VonRoll


Graphic: VonRoll


Graphic: VonRoll

The lecture unit "Wire materials and their processing" provides an overview of the main conductor materials, as well as their manufacture and processing.

- **Types and properties of conductor materials**
- The importance of the fill factor
- Conductor geometries and designs
- Process chain for the production of enameled wire
- Qualification and testing of the wire

Graphic: Aumann

To fulfil the different functions of an electrical winding, a combination of different material is required.

Conductor materials

Graphic: Essex

Tasks:

- Ensure low-loss electrical current flow
- Heat dissipation

Examples:

- Copper
- Aluminum

Insulating materials

Graphic: Heermann

Tasks:

- Separate areas of different electrical potential
- Ensure current flow on the required path

Examples:

- Polyester, polyesterimide lacquer
- Insulating paper
- Thermoplastics, thermosets/duroplastics

Construction materials

Graphic: Risomat

Tasks:

Fixing the shape and position of the winding against:

- forces occurring between the current-carrying conductors and
- potential centrifugal forces in rotating windings

Examples:

- Tubular and cord-shaped cords
- Plastic wires

Copper has established itself in electrical engineering due to its excellent electrical conductivity compared to other metals.

Metal	Relative electrical conductivity [m/(Ω mm²)]	Rel. thermal conductivity [W/mK]	Rel. thermal expansion [10^{-6} K]$^{-1}$	Standardised density [g/cm³]	Standardised costs [€/kg]
Silver	106	108	112	118	8917
Copper	**100**	**100**	**100**	**100**	**100**
Gold	72	76	100	217	633333
Aluminum	62	56	135	30	28
Magnesium	39	41	153	19	42
Zinc	29	29	176	80	36
Nickel	25	15	76	100	208
Cobalt	18	17	71	100	550
Steel	13-17	13-17	71	82	10
Platinum	16	18	53	242	595000
Tin	15	17	124	82	258
Lead	8	9	165	127	32

Source: Copper Institute (as of 2018)

Due to the poor availability and associated higher price of copper, various metal combinations are used in addition to pure materials.

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	Copper	Aluminum	CCA (10%) (copper-clad-aluminum)
Content in the earth's crust	0,006%	7,57%	-
Density	8.92 g/m^3	2.70 g/m^3	3.63 g/m^3
Melting point	1083 °C	660 °C	658 °C
Thermal conductivity	385 W/mK	235 W/mK	240 W/mK
Electrical conductivity - AC	> 58.5 $m/(\Omega mm^2)$	> 35.8 $m/(\Omega mm^2)$	> 37.5 $m/(\Omega mm^2)$
Price (April 2020)	4.5 €/kg	1.3 €/kg	3.6 €/kg - 25.0 €/kg
Price (May 2025)	8,5 €/kg	2.5 €/kg	
Weldability	Good	Bad	Good
Oxidation rate	Low	High	Low
Formability	Good	Flexural strength, long-term flow	Flexural strength

- Conductivity in the outer conductor radius is most important
- Poorer properties of aluminum are improved by the copper coating
- Different thermal conductivities lead to challenges during processing

Sources: ISE, finanzen.net, German Copper Institute, Elektrisola

FAPS

The application requirements determine the choice of conductor material, with aluminum being the most commonly used material after copper.

Wire material	Field of application	Technical background
Aluminum	**Loudspeakers, microphone coils**	- **Weight reduction** - **Material costs**
Brass	Antennas and waveguides, electronics for coils with thin conductor cross-sections	Good strength
Silver	Motors for medical technology, optical industry	- Good strength - Good spring properties - Electrical conductivity
Gold	Antennas, flip-chip and chip-to-chip bonding	Good suitability for contacting
Nickel	Heating applications, Electrical components, Chemical and electrotechnical industry	- Good mechanical properties - Corrosion resistance - High temp. coefficient - High specific electrical resistance
Special wire CCA	High frequency coils	High electrical conductivity with low weight
Copper	**Construction sector (pipework), electronics (motors etc.), electrical engineering and communication (computer chips)**	- **Very good heat conductor** - **Very good electrical conductivity**

Source: Copper Institute (as of 2018)

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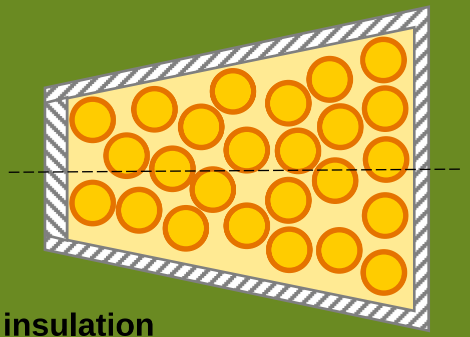
Graphic: Aumann

Fill factor of an electrical winding may be calculated using different methods and attention must be paid to the definition of the fill factor in use.

Slot fill factor

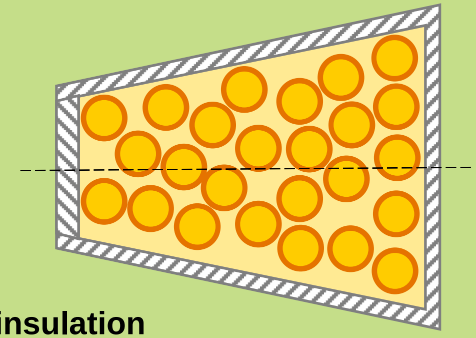
$$k_N = \frac{z_n \cdot \mathbf{d_i^2}}{A_{n\,i}}$$



with
- z_n – Number of wires per slot
- d_i – Outer diameter of **wire and insulation**
- $A_{n\,i}$ – Cross section of the **insulated slot**

Mechanical fill factor

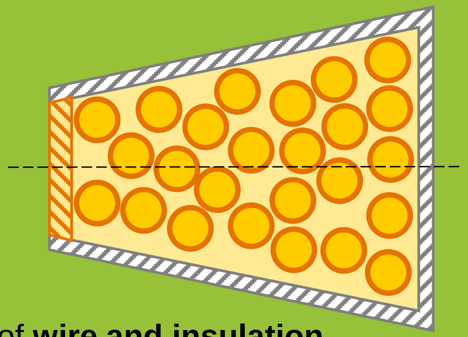
$$k_{mech} = \frac{A_{Wire,i}}{A_{n\,i}} = \frac{\frac{1}{4} \cdot d_i^2 \cdot \pi \cdot z_n}{A_{n\,i}}$$



with
- z_n – Number of wires per slot
- d_i – Outer diameter of **wire and insulation**
- $A_{n\,i}$ – Cross section of the **insulated slot**

Winding fill factor

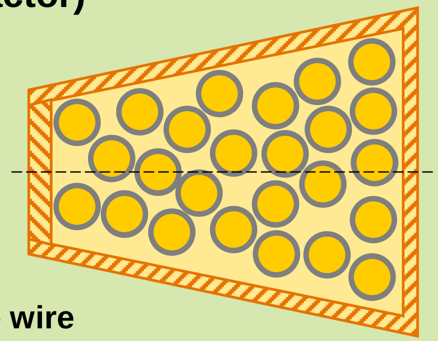
$$k_W = \frac{z_n \cdot d_{i\,max}^2}{A_{n\,eff}}$$



with
- z_n – Number of wires per slot
- $d_{i\,maxx}$ – **Largest** outer diameter of **wire and insulation**
- $A_{n\,eff}$ – Cross section of the insulated slot **and slot opening**

Electrical fill factor (copper fill factor)

$$k_{elec} = \frac{\frac{1}{4} \cdot d_{Cu}^2 \cdot \pi \cdot z_n}{A_n}$$

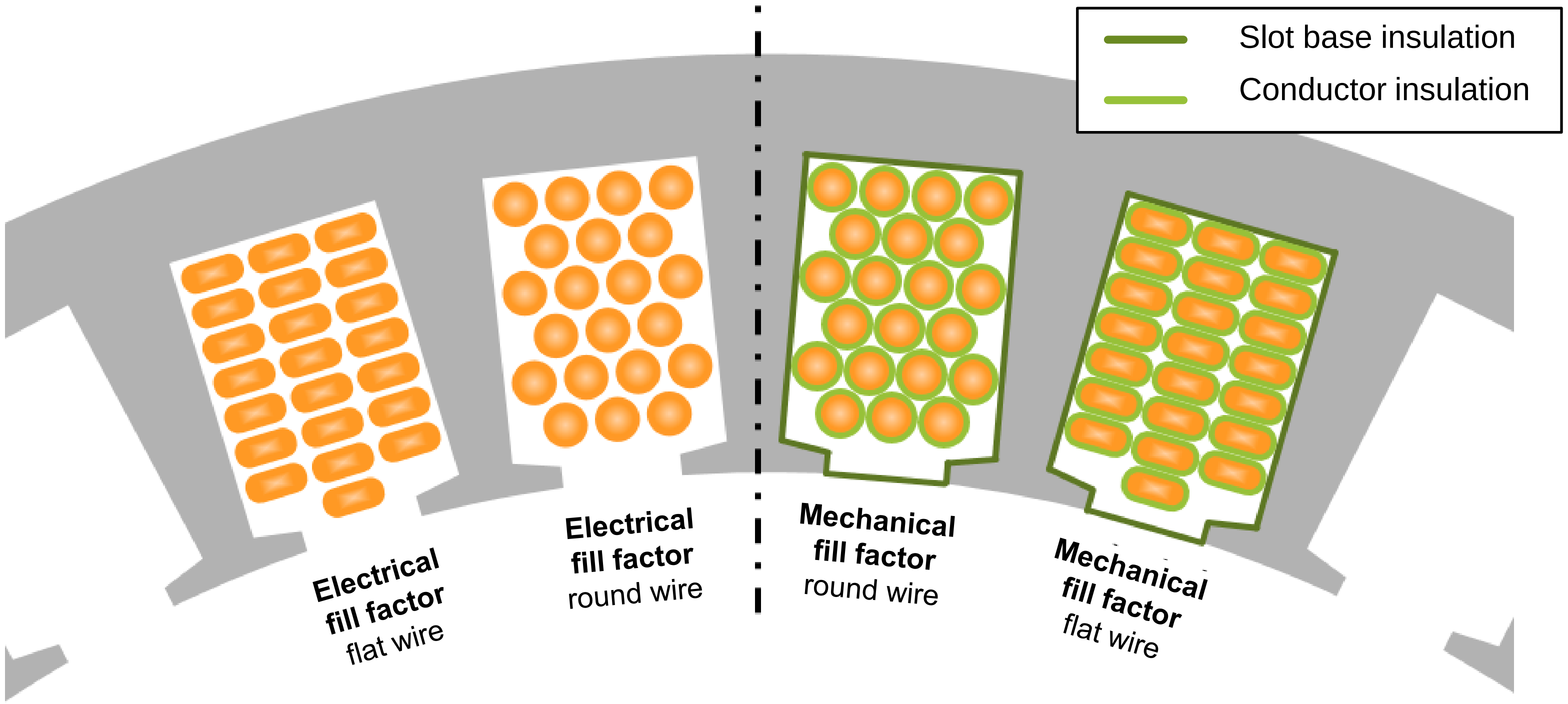


winding
- z_n – Number of wires per slot
- d_{Cu} – Nominal diameter of the **bare wire**
- A_n – **Total** slot cross section

The fill factor is a measure of the amount of conductive material in relation to the area of the slot, regardless of the shape of the conductor.

FAPS

$$k_{el} = \frac{A_l \times z_n}{A_n}$$



$$k_{me} = \frac{A_{li} \times z_n}{A_{ni}}$$

k_{el} :	Electrical fill factor (copper fill factor) [%]	k_{me} :	Mechanical fill factor [%]
A_l :	Cross-sectional area of the bare wire [mm²]	A_{li} :	Cross-sectional area of the insulated wire [mm²]
A_n :	Total slot cross-section [mm²]	A_{ni} :	Slot cross-section of the insulated slot [mm²]
z_n :	Number of wires per slot	z_n :	Number of wires per slot

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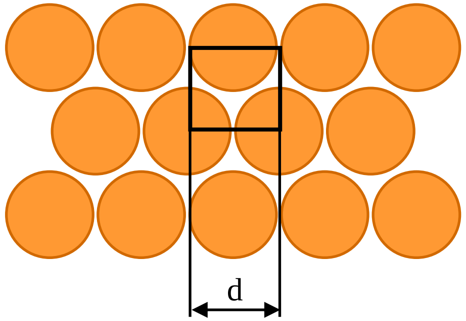
Graphic: Aumann

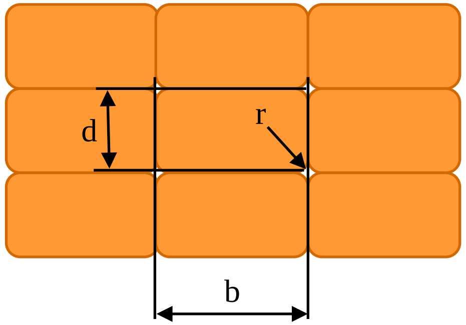
The cross-section of wires used for electrical windings can be round, rectangular, square or polygonal.

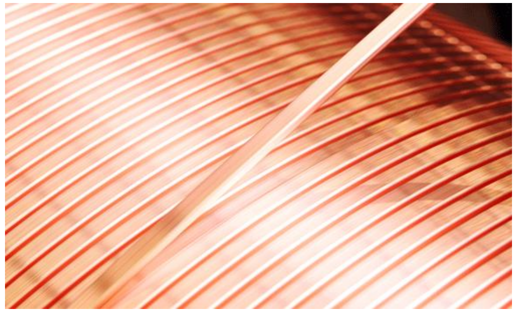
Round wire



Graphic: Bronmetal

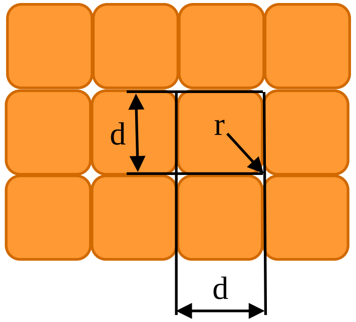
Flat wire





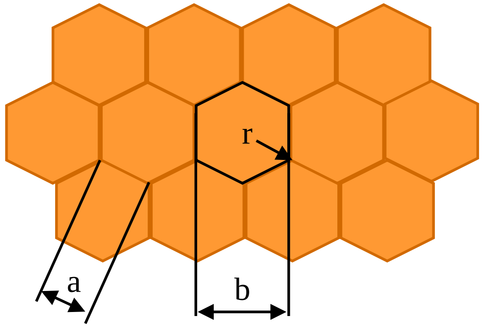
Graphic: SHWire

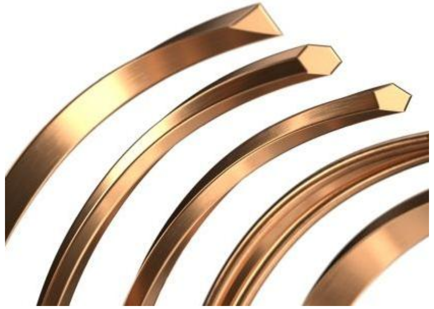
Square wire



Graphic: Elektrisola

Profile wire

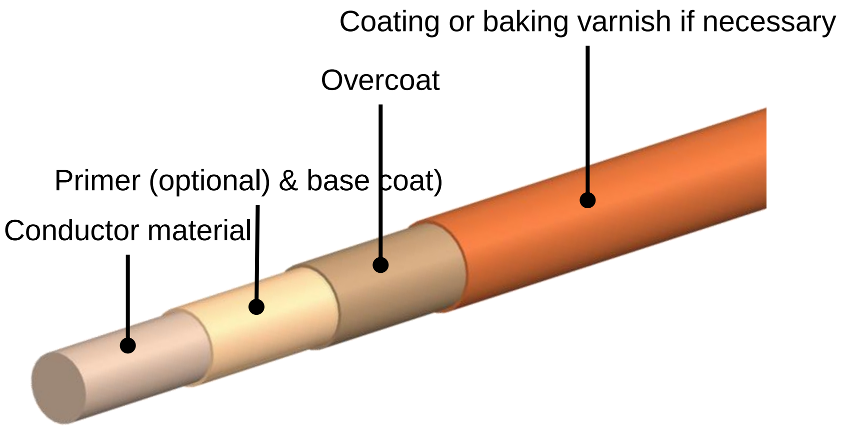




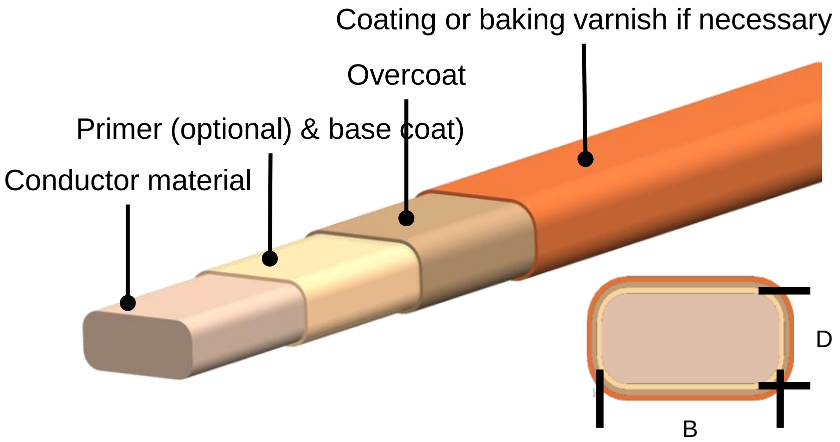
Graphic: Leoni

The structure of an enameled electrical conductor in winding technology differs from the material only in its geometry.

Enameled insulated round conductor



Enameled insulated flat wire

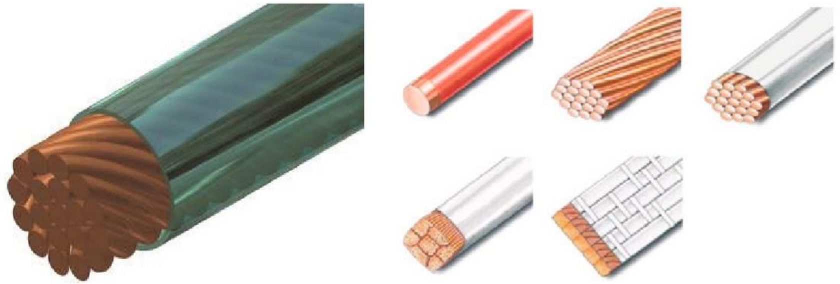


- The carrier material can consist of different materials
- A base coat is applied over the carrier material to ensure defect-free application of the electrical insulation
- After the base coat, the electrical insulation layer (overcoat) is applied
  - Overcoat consists of synthetic resins and/or organic solvents
  - Curing takes place through a baking process
- The insulation is then supplemented with functional layers, such as an adhesive coating or a lubricating layer

In addition to the variation in conductor geometry, many small partial conductors can be combined to form a common conductor bundle, called a stranded conductor or Litz wire.

Characteristic sizes of a stranded conductor

Structure of stranded conductors

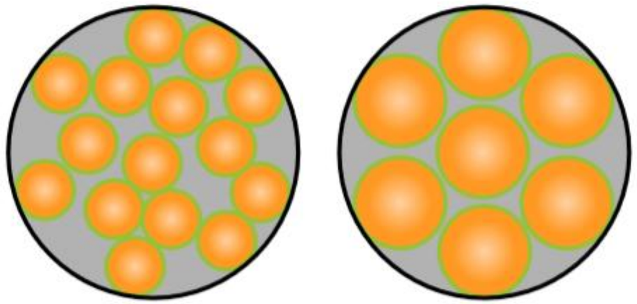


Graphics: Manual of winding technology

- The twisting of different wires creates a "twist lay" which can vary in length and direction.
- To ensure dimensional accuracy, the individual enameled copper wires can be formed into strands with or without a sheath.
- The length after which a single conductor or bundle of conductors has completed a complete circuit around the circumference of the strand is referred to as the lay length.
- The direction of the lay is referred to as the twist direction, with a distinction being made between S-twist and Z-twist.
- Round conductor profiles can also be converted to square or rectangular cross-sections by means of a downstream pressing and roll-forming process in order to increase the packing density in the winding space.

Depending on the number of stranding steps in production, one bundle can be produced in one step or several bundles can be produced for several stranding steps.

Single-level structure

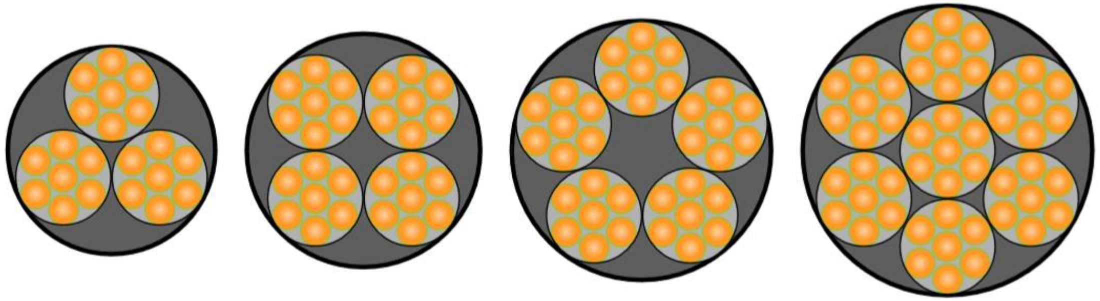


Bunched stranded wire

Concentric stranded wire

Conductor insulation

Multi-level structure



- Depending on the structure of the stranded conductor, a distinction must be made between bunched stranded wire or concentric stranded wire.
  - Bunched strands consist of any number of individual wires which are twisted without a defined geometric position.
  - With concentric strands, the individual wires follow a predetermined geometry in which the individual wires are arranged in a ring around the centre of the cross-section.
- In a multi-level conductor construction, smaller preconstructed strands (bundles) are twisted together to form a larger strand.
- Types of twist configurations are Sz-, Ss-, Zz- or Zs-twists (see previous slide).

Stranded conductors have a wide variety of design parameters, which can significantly influence the properties of the stranded wire.

		Design parameters						
		Conducting material	Wire diameter	Number of conductors	Number of bundles	Lay length	Direction of twist	Increase due to insulation
Litz wire properties	R_{DC}	X	X	X				
	R_{AC}	X	X		X	X		
	J		X	X				
	A_{Ladder} / A_{Litze}		X		X	X	X	X
	U_{BV}							X
	Outer diameter	X	X	X	X	X	X	X
	Dimensional stability		X	X	X	X	X	
	Mech. Flexibility	X	X	X	X	X		
	Flexural fatigue strength	X	X		X	X		
	Tensile strength	X						
	Roundness		X	X	X	X	X	

The stranding process is decisive for the subsequent quality and usability of the stranded conductor.

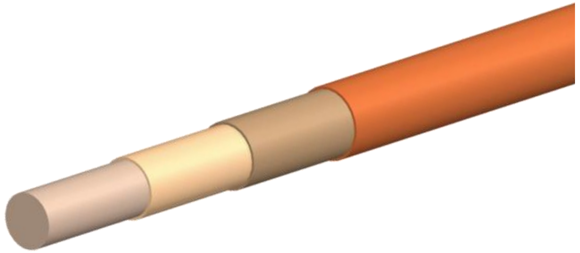
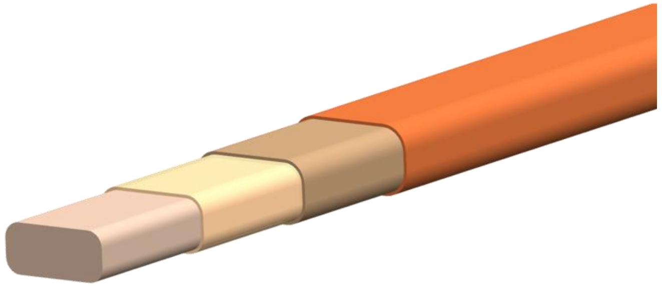
Process

Stranding

Video:
HSM Wire International

https://www.youtube.com/watch?v=_sco8W35cr8

The different conductor topologies offer a wide range of advantages and disadvantages, whose primary characteristics are described below.

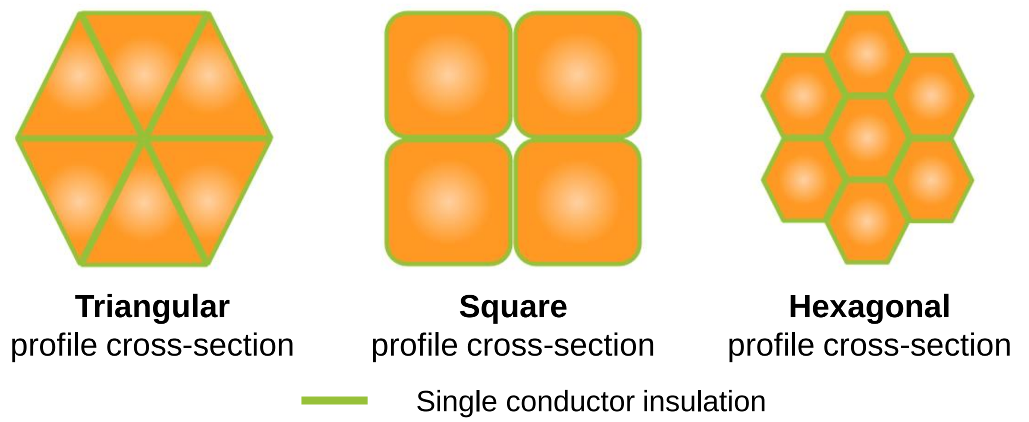
Round wire		Flat wire		Stranded conductor	
Advantages	**Disadvantages**	**Advantages**	**Disadvantages**	**Advantages**	**Disadvantages**
▪ Low current displacement effects (skin, proximity effect) ▪ Flexible processing	▪ Low packing density ▪ Low electrical fill factor	▪ Very high packing density ▪ Very high electrical fill factor	▪ High current displacement effects (skin, proximity effect) ▪ Inflexible processing	▪ Very low current displacement effects (skin, proximity effect) ▪ Very flexible processing	▪ Low packing density ▪ Low electrical fill factor ▪ Difficult to produce
					

Graphics: FAPS

In addition to the "standard conductor geometries" described previously, special geometries are also used depending on the application.

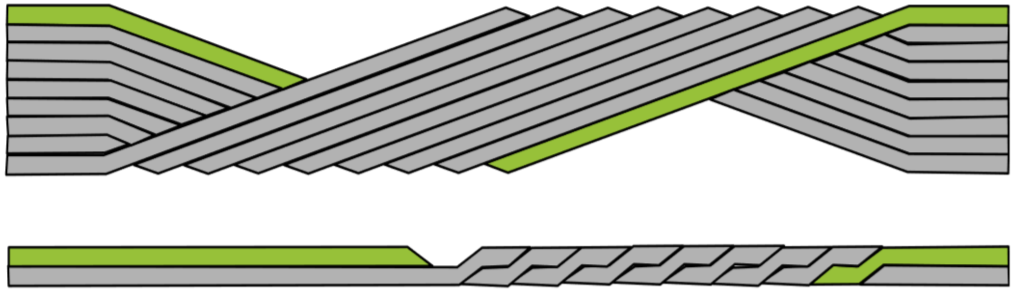
Profile wires

- All cross-sectional geometries that do not correspond to those of round and flat wires are referred to as profile or shaped wires.
- Depending on their geometry, shaped wires are several times more expensive to produce than comparable round wires.
- Their use is limited to special applications and very large sizes.
- The overall objective is to increase the copper fill factor.



Roebel rods (see also lection F6)

- Interlocked and individually insulated shaped rods
- Connected in parallel within a slot
- High production cost due to complex production of the conductors
- Use is limited to large electrical machines with high output (upper kilowatt and megawatt range)



Graphics: FAPS

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Graphic: Aumann

To produce semi-finished copper products, copper must first be extracted from copper ore.

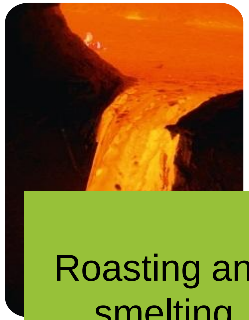
Copper production

Grinding the copper ores

→

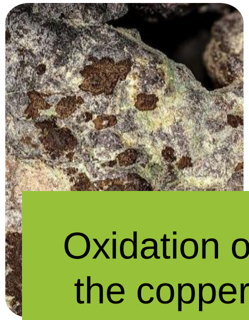
Flotation

→



Roasting and smelting

→



Oxidation of the copper stone

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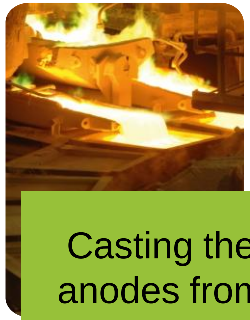


Casting the anodes from blister copper

→

Electrolytic refining

Graphics: German Copper Institute, Mineral Atlas

- After grinding the raw ore, it initially has a copper content of less than **2%.**
- The accompanying rock is separated by flotation (physical-chemical separation process) and the copper content is increased to up to 25%.
- It is then roasted, whereby sulphur dioxide and iron oxide are produced in two process steps with the addition of oxygen.
- After oxidation, the iron and sulphur are bound in a floating slag and then removed by casting → the copper content is now increased to approx. 98%.
- For further enrichment, the raw copper is cleaned in an electrolysis bath. The (electrolyte) copper is deposited on an anode plate → the copper content is now approx. **99.99%.**

To produce semi-finished copper products, copper must first be extracted from copper ore.

Process

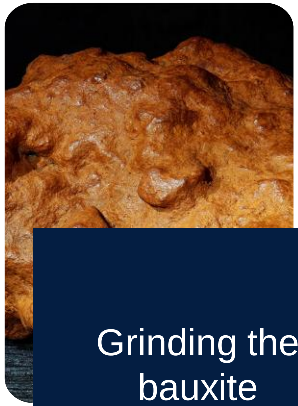
How Copper is made animation

Video:
Karthi Explains

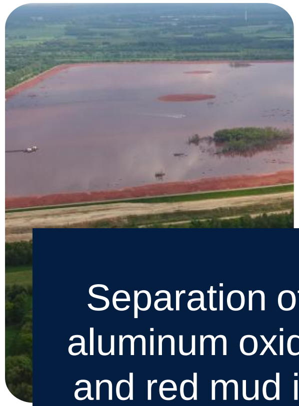
https://www.youtube.com/watch?v=DY7tmUNMydA

In order to achieve the degree of purity of semi-finished aluminum products, several preliminary processes are required, similar to copper production.

Aluminum production



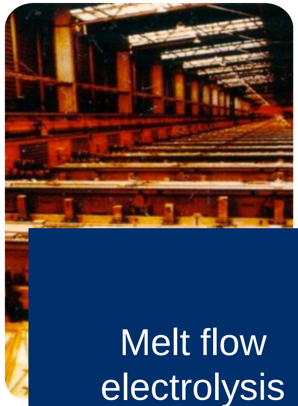
Grinding the bauxite



Separation of aluminum oxide and red mud in an autoclave

Filtering and stirring the aluminum oxide

Adding cryolite to reduce the melting temperature



Melt flow electrolysis

Images: Mineral Atlas, Rosiwal

- The alumina is dissolved from ground bauxite using the Bayer process.
- By heating in an autoclave (160°C - 270°C) with the addition of a digestion liquor (Na_2O), the aluminum oxide dissolves as sodium aluminate and can thus be separated from the remaining components (the so-called red mud)..
- The dissolved sodium aluminate is concentrated and filtered. Aluminum hydroxide is then stirred out. By heating to 1200°C - 1300°C, aluminum oxide (also known as alumina in this form) can be obtained with a relatively high degree of purity.
- The second step is the reduction of the aluminum oxide to aluminum (purity 99.9%) by fused-salt electrolysis at 950°C, whereby the melting point of the aluminum oxide (2050°C) is significantly reduced by the addition of cryolite.

Further intermediate steps are required in copper and aluminum production to manufacture round and flat wire from sheets.

→ Semi-finished products in wire production are also called **wire rods or drawing stock.**

→ Standardized for copper (DIN EN 1977) and aluminum (DIN EN 1715) with regard to dimensions and properties.

Copper wire rods

- Dimensional range from 6.00 mm to 35.00 mm (PREFERRED: extruded material of 7-8 mm)
- Four possible manufacturing processes: continuous casting and hot rolling, continuous or semi-continuous casting and cold rolling, rolling of wire billets or square billets, and extrusion moulding
- Depending on the process, a subsequent annealing process in an inert environment is required
  → Ensuring the necessary elongation at break for the subsequent drawing or rolling process

Aluminum wire rods

- Dimensional range from 7.00 mm to 32.00 mm
- Two manufacturing processes:
  - Hot rolling
  - Extrusion (extrusion is used particularly for high-alloy aluminum materials and small batch sizes)

The basis for the further processing of round and flat wires is the is the extruded wire or extruded wire rod.

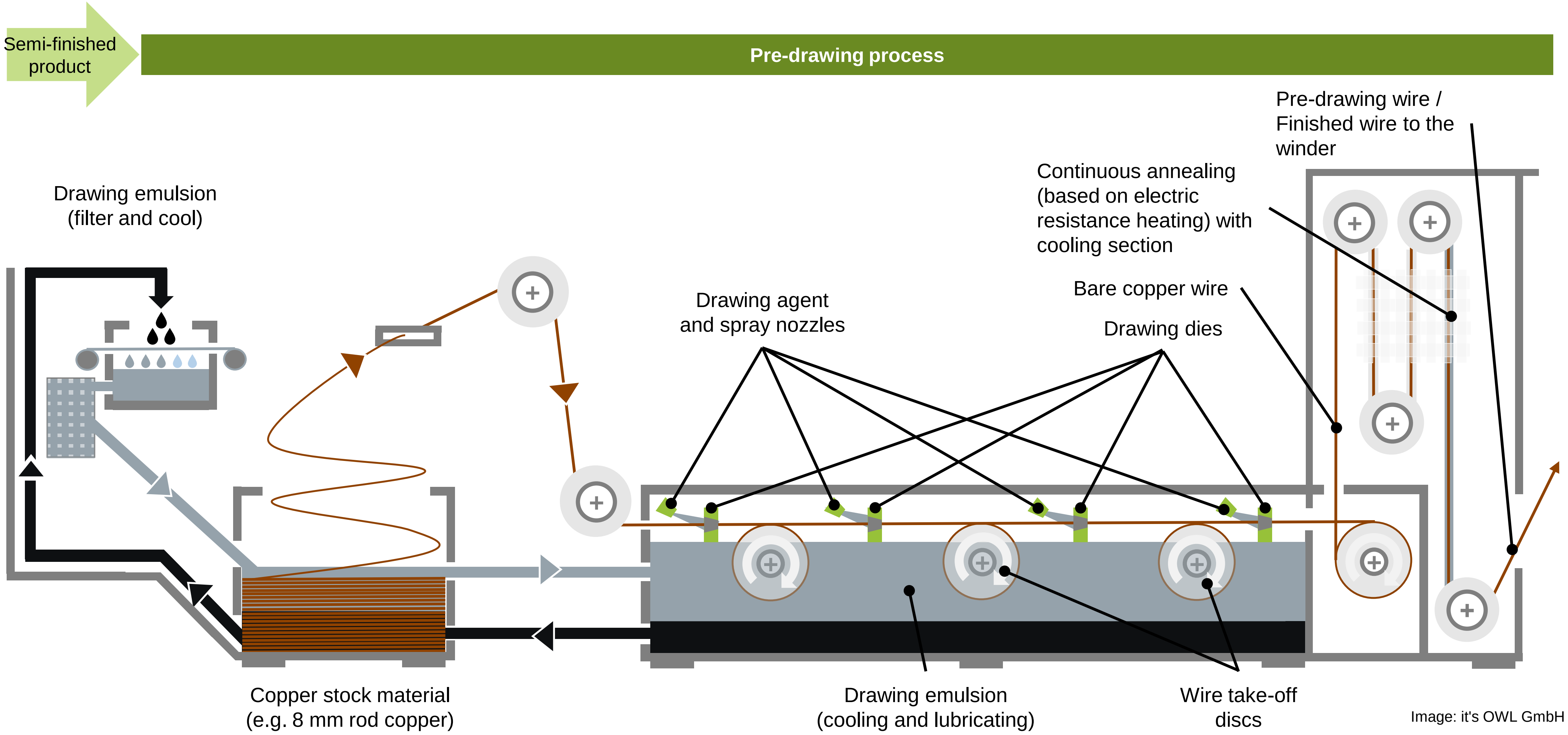
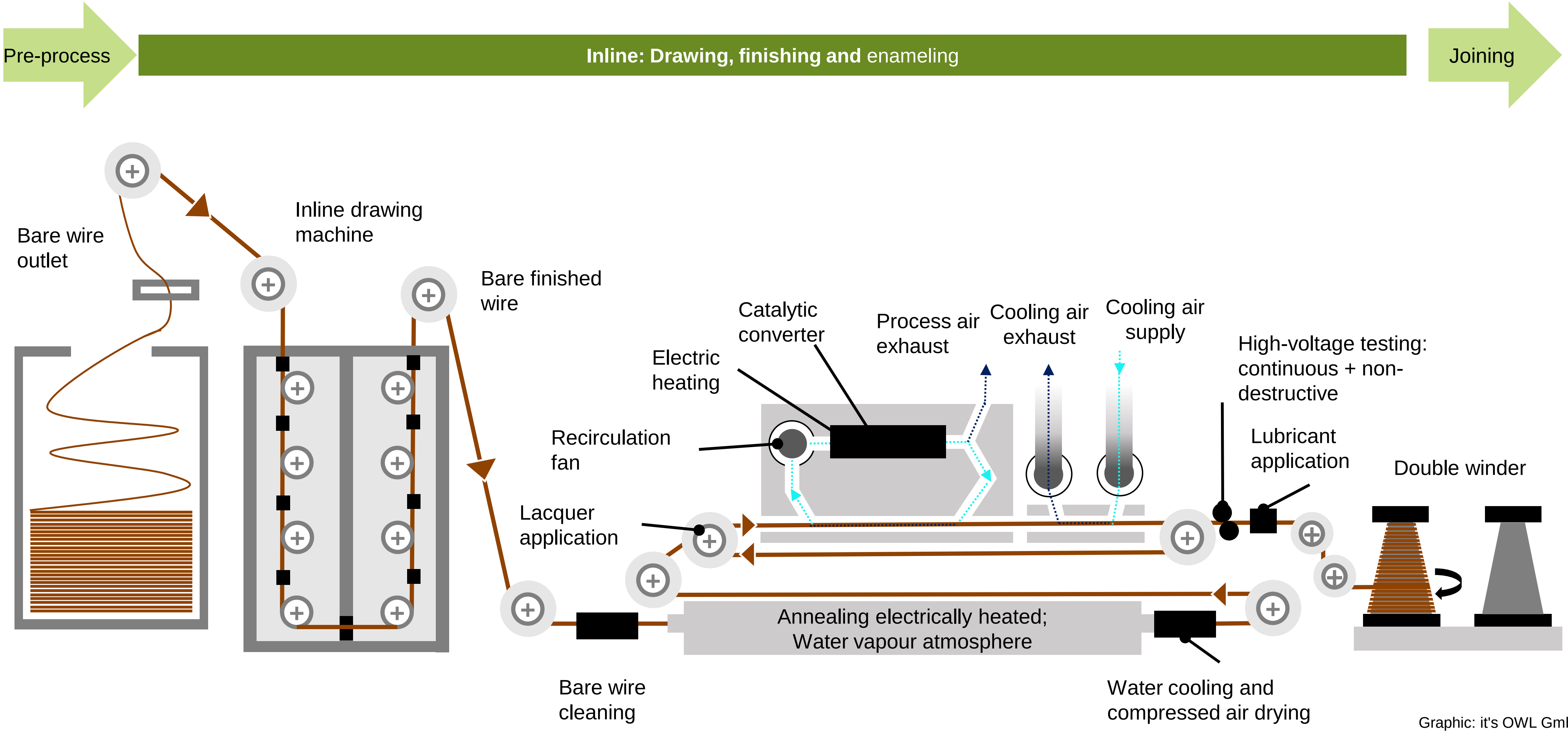


Image: it's OWL GmbH

The state of the art for round and flat wires is wire drawing using an inline drawing process, in which forming, finishing and enameling take place without interruption.



Graphic: it's OWL GmbH

The state of the art for round and flat wires is wire drawing using an inline drawing process, in which forming, finishing and enameling take place without interruption.

Process

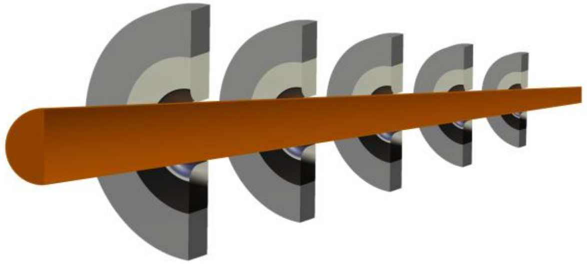
Copper production process of enamel coated wire

Video:
Condumex Cables

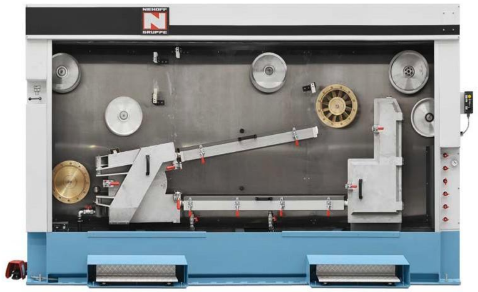
https://www.youtube.com/watch?v=hRCxdCJ_Z2s

The state of the art for round and flat wires is wire drawing using an inline drawing process, in which forming, finishing and enameling take place without interruption.

Wire drawing to the desired diameter → Soft annealing of the wire → Cleaning



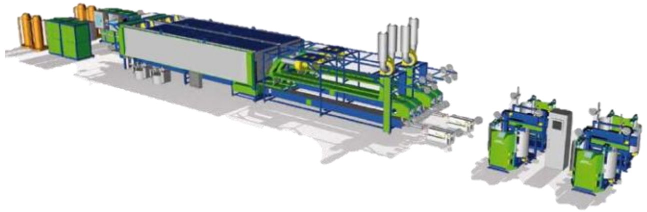
Graphic: FAPS



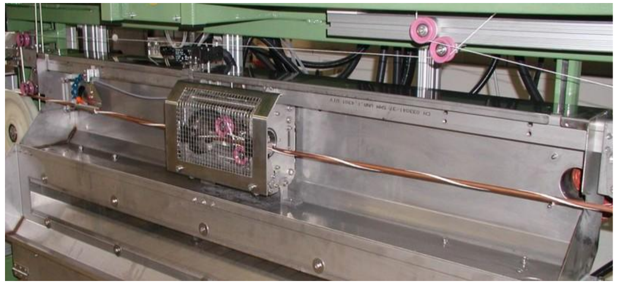
Graphic: Niehoff GmbH

Graphic: GEO-Reinigungstechnik GmbH

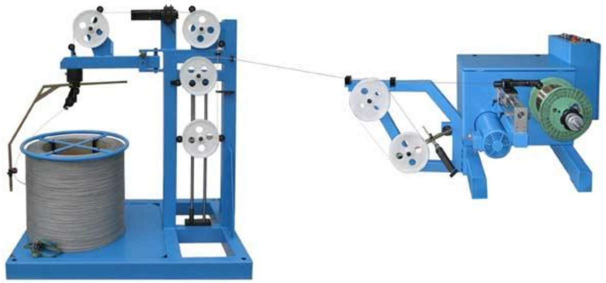
Application of several layers of insulation → Application of lubricant / functional layer → Winding the wire onto a storage spool



Graphic: Essex



Graphic: Boockmann GmbH



Graphic: Mobac GmbH

The wire is drawn to the desired cross section with the aid of coolant and lubricant through dies of decreasing diameter to the desired cross section.

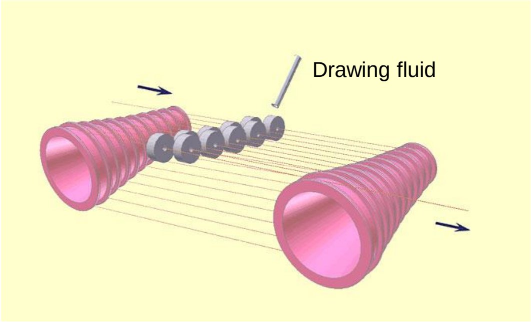
Drawing

- Maximum degree of forming per drawing stage φmax = 0.2 to φmax = 0.5
- Compensation of the increase in length due to the cross-sectional tapering is required
- Horizontal and vertical design possible using drawing cones or drawing discs

Drawing disc and drawing cone

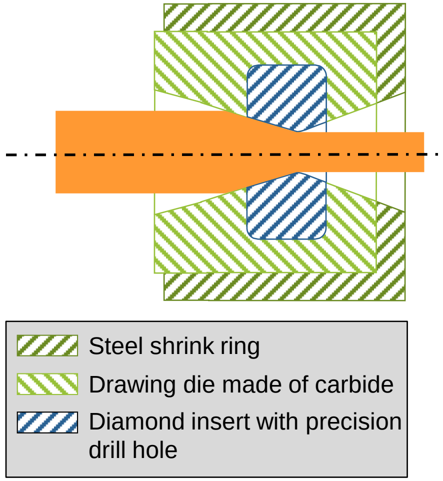
Graphic: Hofmann Ceramic GmbH

2 cones with 10 drawing stages



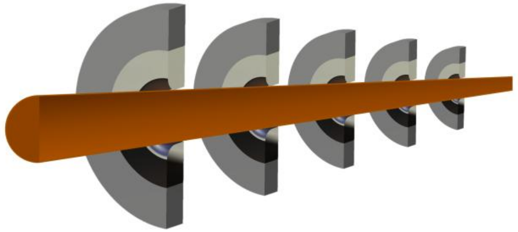
Graphic: Elektrisola

Drawing die assembly



Graphic: FAPS

5 Drawing stages with dies



Graphic: FAPS

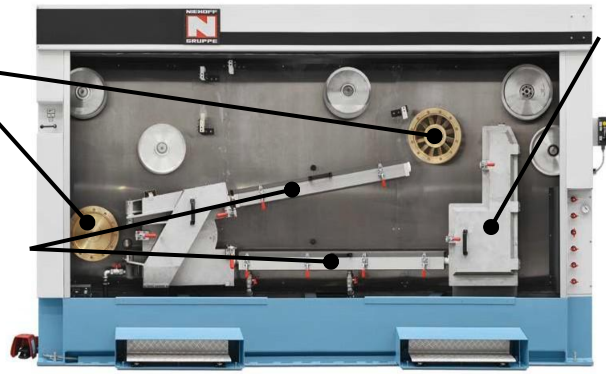
For further processing or to restore high conductivities it may be necessary to reduce the dislocation density by soft annealing.

FAPS

Annealing

- Heating by means of electric current
- Heating by induction
- Inert environment to avoid oxides
- Copper: 400 °C - 500 °C
- Aluminum: 300 °C - 400 °C
- Temperature holding time depends on wire thickness and desired elongation at break

Continuous resistance annealing



Graphic: Niehoff GmbH

Material: Wire made of Cu-ETP	Yield strength $R_{p0,2}$ [N/mm²]	Tensile strength R_m [N/mm²]	Electrical con-ductivity at 20° [m/Ωmm²]
Soft annealed (Brinell hardness HV=55)	max. 120	min. 200	0,01724
Cold drawn (Brinell hardness HV=80)	min. 200	min. 250	0,01754
Cold drawn (Brinell hardness HV=95)	min. 260	min. 300	0,01754

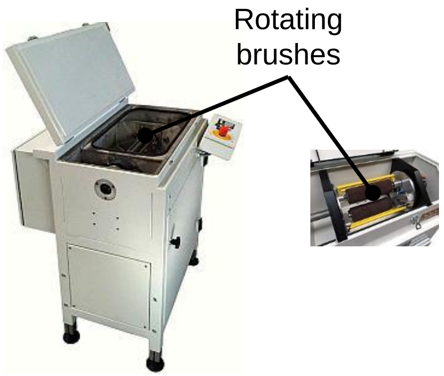
Source: German Copper Institute

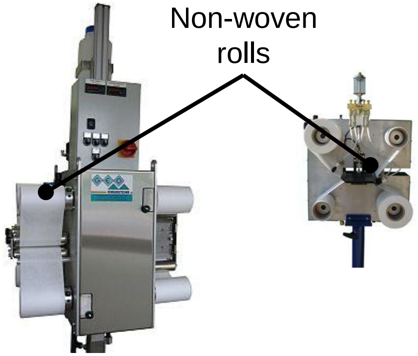
To achieve good conditions for the adhesion of insulating varnish, the wire surface must be free of impurities.

Cleaning

- Essential for optimum adhesion of the enamel to the wire
- Cleaning is critical for windings subjected to high thermal and mechanical stresses
- Coolants and lubricants burn off during annealing
- Non-contact with ultrasonic modules, high pressure jets or steam jets
- Mechanical with brushes, cloths, belts or sponges

Contactless cleaning		Mechanical cleaning	
Ultrasonic module	High pressure cleaning	Cleaning with brushes	Cleaning with fabrics



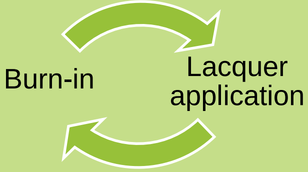


Graphics: Geo-cleaning technology

The insulating varnish is applied in over to thirty layers depending on the desired thickness on the respective conductor.

Coating process

A distinction is made between two different procedures differentiated:



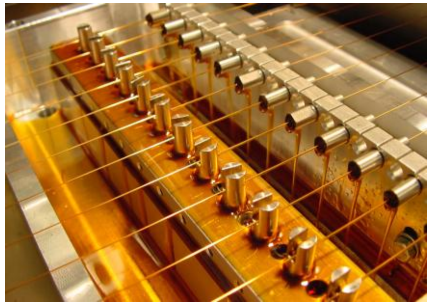
- Nozzle coating application
- Lacquer application with felt
  - 12 - 24 layers for thinner wires (0.012mm to 0.03mm)
  - 15 - 30 layers for medium wires (0.04mm to 0.15mm)
  - 12 - 24 layers for thicker wires (0.15mm to 5mm)

Structure of the coating system depending on the wire diameter:

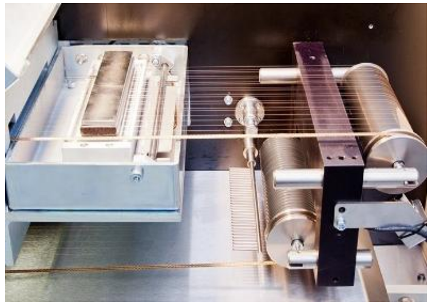
- Vertical enameling system for larger wire diameters and profile wires
- Horizontal enameling system for thin wires

Classification of wire insulation according to DIN IEC 60317:

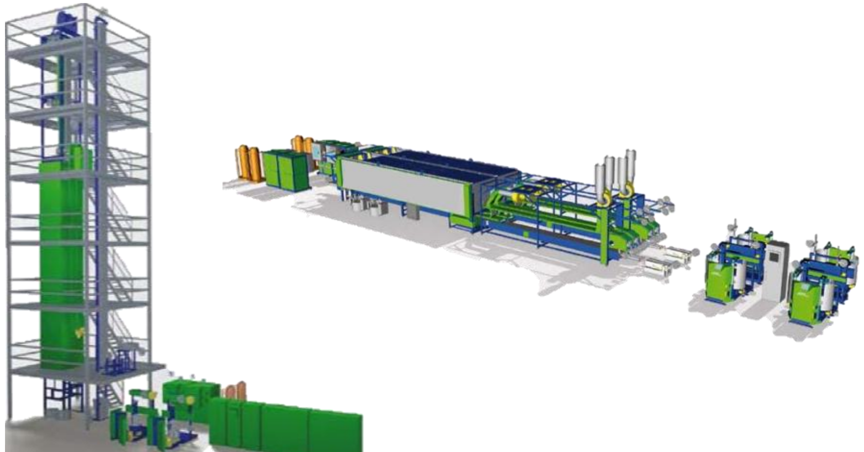
- Grade 1:
  ~ 60 µm for flat conductors / round wires according to table
- Grade 2:
  ~ 120 µm for flat conductors / round wires according to table
- Grade 3:
  ~ 180 µm for flat conductors / round wires according to table



Graphic: Aumann



Graphic: Aumann



Graphic: Essex

The video demonstrates nozzle-application of the insulation layer onto the copper conductor.

Process

Nozzle coating

Video:
Aumann Espelkamp
Ltd.

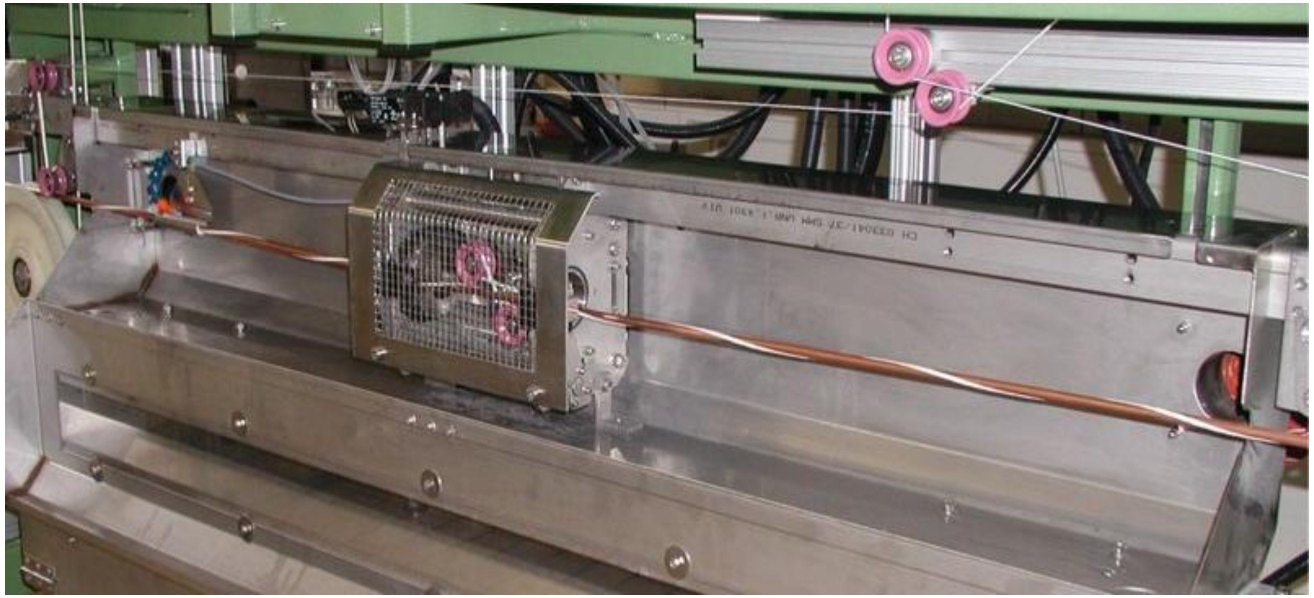
https://www.youtube.com/watch?v=dMaEu3vMlkUtch?v=hRCxdCJ_Z2s

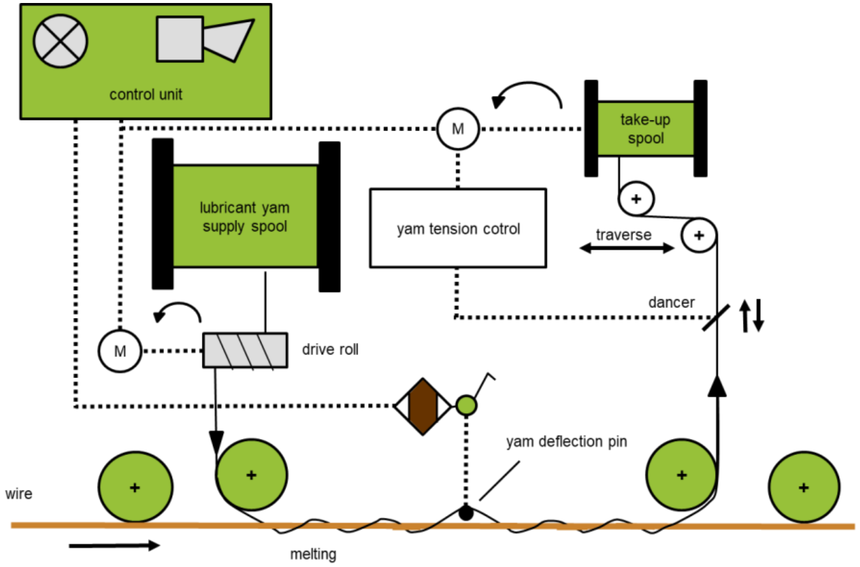
Due to the frictional properties of the insulation, further processing of enameled wires requires the application of lubricants to the wire.

Lubricant application

To reduce surface friction, a thin layer of lubricant is applied after each painting process:

- Consists of paraffin or a dissolved form of beeswax
- Light petrol solutions are used to dissolve the solids
- Applied using impregnated textile thread
- Serves to ensure manageability during unwinding and winding, wire guidance in the machine and in the winding process





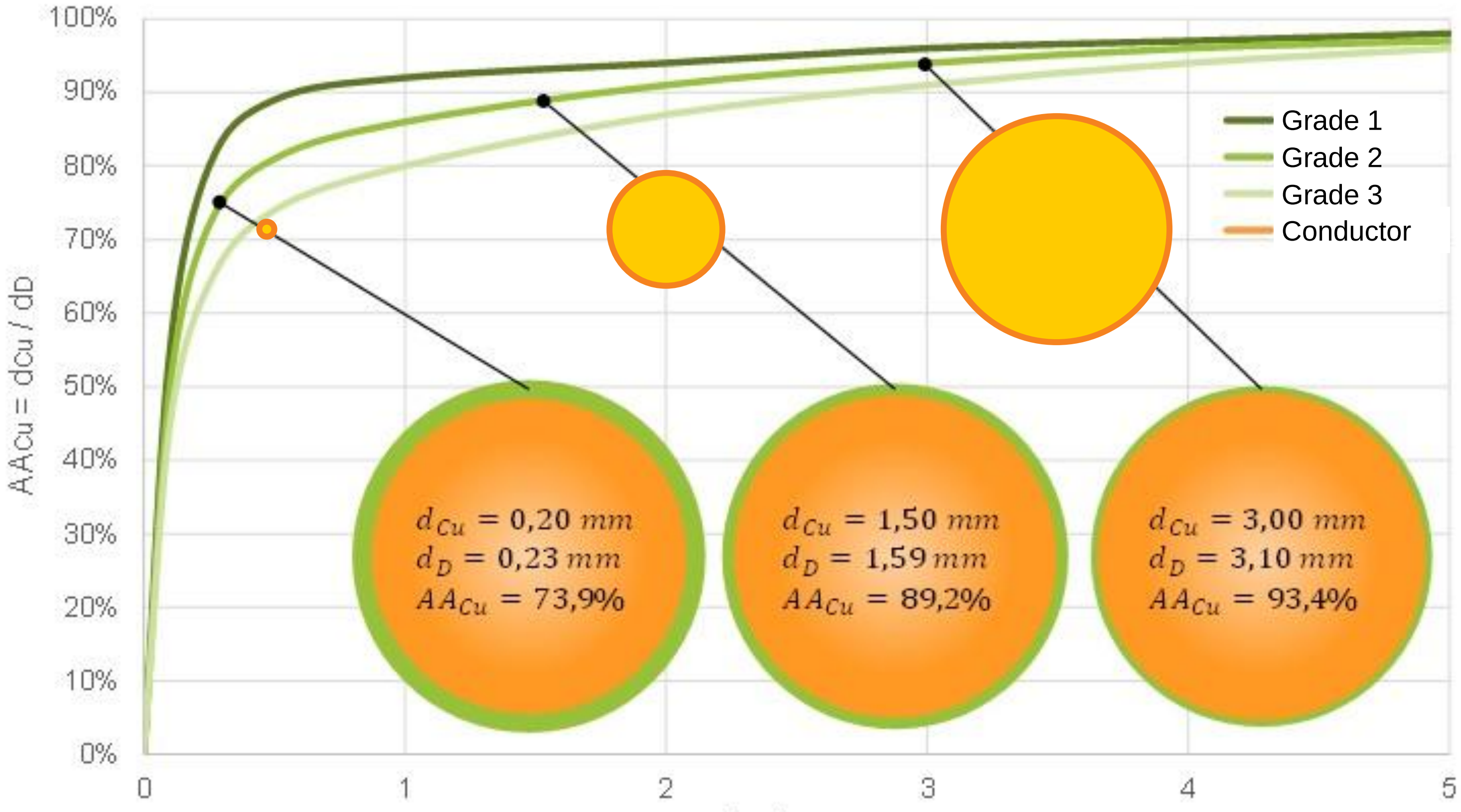
Graphics: Boockmann GmbH

The thickness of the enameled insulation layer on the wires influences the insulation strength, which prevents voltage breakdowns.

Importance of enameled insulation

- A key parameter in the qualification of the insulation layer is the breakdown voltage. It is a measure of the insulation strength of insulating materials.
- If an insulating material is contacted by two electrodes and an increasing voltage is applied, a breakdown occurs when the insulation strength is reached.
- Breakdown voltage of enameled copper wire depends on the thickness and the uniform application of the enamel layer. The surface quality of the bare wire and the degree of baking of the enamel layer are further influencing factors.

Degree of insulation: Ratio of the coating thickness to the copper thickness



with

- d_{Cu} - Nominal conductor diameter (bare wire)
- AA_{Cu} - Proportion of copper in the total cross-section
- d_D - Diameter of the insulated wire

Graphic: FAPS

The different degrees of insulation have a direct influence on the resulting diameter of the conductor and the number of allowable faults in the insulation.

FAPS

Insulation grade 1			Insulation grade 2			Insulation grade 3		
Copper diameter [mm]	Min. increase due to insulation [mm]	Max. total diameter [mm]	Copper diameter [mm]	Min. increase due to insulation [mm]	Max. total diameter [mm]	Copper diameter [mm]	Min. increase due to insulation [mm]	Max. total diameter [mm]
0,355	0,020	0,392	0,355	0,038	0,411	0,355	0,057	0,428
0,500	0,024	0,544	0,500	0,045	0,566	0,500	0,067	0,587
1,00	0,034	1,062	1,00	0,063	1,094	1,00	0,095	1,124
2,00	0,040	2,074	2,00	0,075	2,112	2,00	0,113	2,148
3,00	0,045	3,083	3,00	0,084	3,126	3,00	0,127	3,166
4,00	0,047	4,088	4,00	0,089	4,133	4,00	0,134	4,176
5,00	0,050	5,093	5,00	0,094	5,141	5,00	0,142	5,186

The higher the degree of insulation, the higher the minimum breakdown voltage and the lower the risk of pinholes in the insulation.

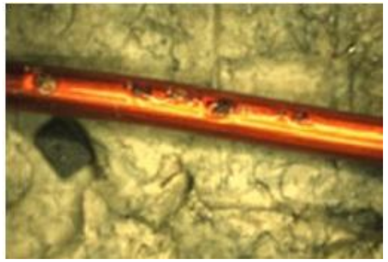
Maximum number of pinholes per 30 m			
Nominal diameter of the conductor [mm]	**Grade 1**	**Grade 2**	**Grade 3**
Up to 0.050	60	24	-
0,050 - 0,080	60	24	3
0,080 - 0,125	40	15	3
0,125 - 1,600	25	5	3

Source: DIN IEC 60317

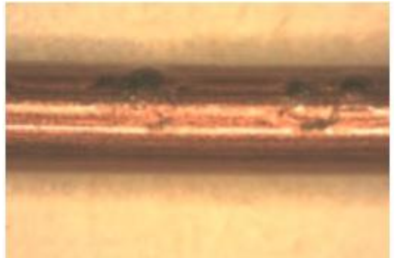
Various defect patterns can occur during painting, which have different causes.

Defect	Cause
Bubbles on the wire surface	Contamination of the surface before painting (solvent or moisture)
Poor insulation adhesion: Rough surface	Process speed too high Temperature too low Worn felt applicator / incorrect nozzle selection
Incorrect elongation at break	Annealing temperature too low / high
Incorrect frictional properties: problems when stripping	Problems with lubricant application
Faulty insulation: negative effects on partial discharge values and dielectric strength	Example: Paint removal by sanding on roller / machine elements Insufficient paint drying

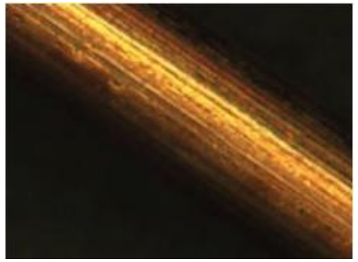
Contamination of the bare wire



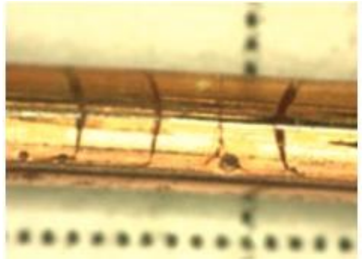
Bubble formation



Groove formation



Cracking



Graphics: Manual of winding technology

As varnish insulation coatings are not suitable for every area of application, materials from the textile industry with various additives are also used.

Generally speaking, varnishes differ as follows:

- Solderable varnish, such as polyurethane
- Heat-resistant varnish, such as polyester
- Solvent-resistant varnish, s such as polyesteramidimide
- Baking varnishes, such as polyesteramide

Other insulation materials are

- Nylon
- Textile insulation for wires, such as E-glass yarn (alkali-free and impregnated with varnish)
- Tape and foil insulation, such as mica / polyester film or tape with epoxy resin

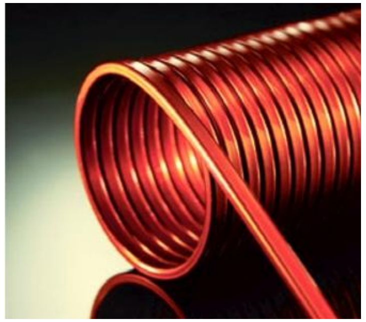
Material	Temperature index	Thermal shock	Softening temperature	Tin plating	Chemical resistance
Polyvinyl acetal	120°C	155°C	>215°C	No	Very good
Polyurethane	180°C	200°C	>250°C	Yes	Good
T. Poly. (imide) + polyamideimide	200°C	220°C	>340°C	No	Good
Polyamideimide	220°C	240°C	>400°C	No	Very good
Polyimide	240°C	260°C	>500°C	No	Very good

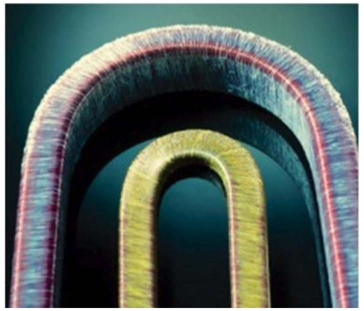
Source: Essex

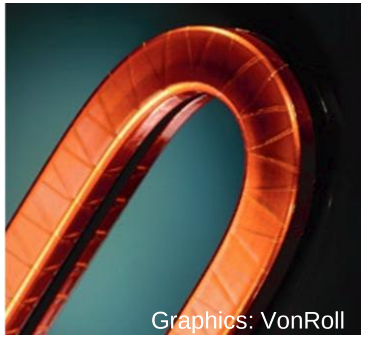
FAPS

The different insulation materials and substances lead to a wide variety of wire applications.

Application	Enameled insulated round wires	Enameled insulated flat wires	Enameled stranded wires	Glass yarn insulated wires	Tape and foil insulated wires
Transformers	X	X	X		
Power tools	X				
NV motors (industry)	X			X	X
NV engines (automotive)	X	X			
Electrical industry	X		X		
Traction motors	X	X		X	X
Wind turbine generators			X	X	X
High-voltage rotary machines			X	X	X
Large generators				X	
Repair industry		X		X	X







Graphics: VonRoll

The lecture unit "Wire materials and their processing" provides an overview of the main conductor materials, as well as their manufacture and processing.

- Types and properties of conductor materials
- The importance of the fill factor
- Conductor geometries and designs
- Process chain for the production of enameled wire
- **Qualification and testing of the wire**

Graphic: Aumann

There are various standards for the qualification and testing of winding wires and their insulations.

Lecture unit 10

DIN EN 60317-0-1 (Technical delivery conditions) for round wire and DIN EN 60317-0-2 for flat wire include, for example

- Limiting dimensions of conductor cross-sections including various degrees of insulation
- Minimum and maximum values of conductor resistance
- Minimum value of elongation at break
- Springback angle after wire bending
- Minimum breakdown voltage for various ambient temperatures

DIN EN 60851-1 Winding wires - Test methods

- Part 1: General (e.g. areas of application, terms, etc.)
- Part 2: Determination of dimensions (e.g. test equipment for round and flat wires as well as for stranded conductors)
- Part 3: Mechanical properties (e.g. testing elasticity and adhesion, abrasion resistance)
- Part 4: Chemical properties (e.g. resistance to solvents, solderability)
- Part 5: Electrical properties (e.g. breakdown voltage, integrity of the insulation)

The contents of the lecture unit can be further deepened with the following specialized literature:

Essays

- Bickel, B.; et al. (2014): "Analysis of the optimisation potential for increasing the copper fill factor in electrical machines". Journal article
- Riedel, A.; et al. (2019): Calculation of the Copper Filling Factor of Electric Traction Drives including Graphical Representation". Conference paper, DOI 10.1109/ICEMS.2019.8922428

Dissertations

- Wenger, U. *Process optimisation in winding technology through innovative mechanical engineering and control technology approaches.* Dissertation. Bamberg: Meisenbach, 2004. production technology Erlangen. 147. ISBN 3-87525-203-9
- Dobroschke, A. *Flexible automation solutions for the production of winding technology products.* Dissertation. Bamberg: Meisenbach, 2011. production technology Erlangen. 219. ISBN 3-87525-317-5
- Kühl, A. *Flexible automation of stator assembly with the aid of universal ambidextrous kinematics.* Dissertation. Bamberg: Meisenbach, 2014. production technology Erlangen. 251. ISBN 978-3-87525-367-2

The contents of the lecture unit can be further deepened with the following specialized literature:

Textbooks

- Bălă C.; et al. (1976): "Handbuch der Wickeltechnik elektrischer Maschinen", VEB Verlag Technik, Berlin.
- Jordan, W.; (2016): "Technologie kleiner Elektromaschinen Teil 1-4", Technoexpert, Dresden, ISBN 978-3-00-053628-1
- Much W.; et al (1986): "Wicklungen und Montage rotierender elektrischer Maschinen", VEB Verlag Technik, Berlin, ISBN 3-341-00020-8.
- Tzscheutschler R.; et al. (1990): "Technologie des Elektromaschinenbaus", VEB Verlage Technik, Berlin, ISBN 3-341-00851-9

The contents of the lecture unit can be further deepened with the following specialized literature:

Graphics

- AUMANN AG. Wire production [online]. Wire enamelling [Accessed on: 12 August 2021]
- AURUBIS. Developments on the copper market, 8 February 2019, (Copper mail no. 168)
- BECKMÖLLER, S. Winding wires - a high-tech product - possibilities and limits -. Electrical engineering - Current developments in winding technology
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- GEO REINIGUNGSTECHNIK GMBH. Products. Wire cleaning / parts cleaning / deburring [online]
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- Risomat GmbH & Co KG. Processes [online]. Processes and machine concepts according to requirements
- Rosiwal, S. Lecture notes for the lecture Materials Science. Erlangen
- Schwering&Hasse Elektrodraht. Expert system for configuring processes in winding wire production - ExKoP, 6 August 2015
- Schwering & Hasse Elektrodraht GmbH. SHWire Our solutions [online]
- Schorn, S. Mineralienatlas - Fossilienatlas [online]
- vonRoll. Winding wires and strands

FAPS

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THANK YOU